

## Unit 8 – Inference for Categorical Data: Chi-Square

### Notes 2 – Chi-Square Test for Homogeneity

#### Chi-Square or Two Proportions?

- The two proportion z test allow us to compare the proportions of successes between two populations (or treatments) with two proportions
- The chi-square test for homogeneity will allow us to compare the proportion of successes across several populations (or treatments)

#### The Chi-Square Test for Homogeneity

Ex) After turning the age of 18, do males and females differ in their living arrangements with their parents? We select two separate random samples of 200 males and 150 females, who are between the ages of 18-20, to determine where they live. The results are in the table to the right.

|        | With Parents | In Apartment | Dormitory | Total |
|--------|--------------|--------------|-----------|-------|
| Female | 61           | 29           | 60        | 150   |
| Male   | 60           | 27           | 113       | 200   |
| Total  | 121          | 56           | 173       | 350   |

#### Hypotheses

|                                                                                                             |                 |
|-------------------------------------------------------------------------------------------------------------|-----------------|
| To perform a test, we use the following hypotheses, in context:                                             | In our example: |
| $H_0$ : There is _____ in the distribution of a categorical variable for several populations or treatments. | $H_0$ :         |
| $H_a$ : There is _____ in the distribution of a categorical variable for several populations or treatments. | $H_a$ :         |

#### Expected Counts

- We compare the observed counts in a two-way table with the counts we would expect if  $H_0$  were true.
- To find the expected counts, we start by assuming that  $H_0$  is true. In our example, we are assuming there is no difference between males and females in their living arrangements.
- We would expect this proportion to be the same across all living situations. THIS DOES NOT MEAN THE EXPECTED COUNTS WILL BE THE SAME.

Finding expected counts: The expected count of any cell in a two-way table can be found using the following formula:

| Observed Counts |                         |                         |                  |              | Expected Counts |                         |                         |                  |
|-----------------|-------------------------|-------------------------|------------------|--------------|-----------------|-------------------------|-------------------------|------------------|
|                 | <i>With<br/>Parents</i> | <i>In<br/>Apartment</i> | <i>Dormitory</i> | <i>Total</i> |                 | <i>With<br/>Parents</i> | <i>In<br/>Apartment</i> | <i>Dormitory</i> |
| <i>Female</i>   | 61                      | 29                      | 60               | 150          | <i>Female</i>   |                         |                         |                  |
| <i>Male</i>     | 60                      | 27                      | 113              | 200          |                 |                         |                         |                  |
| <i>Total</i>    | 121                     | 56                      | 173              | 350          | <i>Male</i>     |                         |                         |                  |

### Conditions

- Random – Data should be collected using a stratified random sample or randomized experiment
- Independent – When sampling without replacement, check the 10% condition for the table total
- Large Counts – All expected counts across every cell must be greater than 5

### Chi-Square Statistic & P-value

When we have expected and observed counts, we can calculate our familiar chi-square statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

This time, the sum is over all cells (not including the totals!) in the two-way table. The degrees of freedom, because we have two categories now, are

df = \_\_\_\_\_

To find the p-value, use the chi-square statistic you found along with your degrees of freedom:

P-value =  $\chi^2\text{cdf}(\chi^2, 1e99, \text{df})$  on the TI-84

### Example

After turning the age of 18, do males and females differ in their living arrangements with their parents? We select two separate random samples of 200 males and 150 females, who are between the ages of 18-20, to determine where they live. The results are in the table below.

|        | With Parents | In Apartment | Dormitory | Total |
|--------|--------------|--------------|-----------|-------|
| Female | 61           | 29           | 60        | 150   |
| Male   | 60           | 27           | 113       | 200   |
| Total  | 121          | 56           | 173       | 350   |

Do the data suggest, at the 5% significance level, that males and females differ in their living arrangements?